MOHIT KUMAR

Data Scientist

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SUMMARY

Data Scientist with 4.5+ years of experience building and deploying production-grade machine learning systems on GCP. Specialized in regression, churn modeling, explainable AI, and LLM-based systems including Text2SQL and RAG architectures. Proven track record of delivering measurable business impact, including 15% predictive accuracy improvement and $120K+ annual cost savings.

EXPERIENCE

Data Scientist

XPO

- Developed a **Text2SQL Gemini LLM-chatbot** with **Database-Augmented Generation**. Enabled **domain-specific SQL** generation using **Pydantic** schemas
- Built a **semantic-layer filtering** mechanism to reduce irrelevant LLM context injection and improve query precision
- Developed an LLM-based workflow for email classification and automated responses with intelligent document retrieval. Deployed serverlessly on **Google Cloud Functions** with **Pub/Sub**-based event processing, handling **~10,000 emails per day**
- Engineered compound features and trained XGBoost regression model (Optuna tuned), achieving **15% improvement in exact-date ETA prediction accuracy**. Performed cross-validation and feature importance analysis to ensure model robustness
- Led a two-tiered churn modeling framework (classification + revenue regression) achieving **50% churn revenue capture by targeting top 20% customers**. Evaluated using precision-recall tradeoff to optimize business ROI
- Applied **SHAP explainability** techniques to identify churn drivers and support business decision-making

Data Science Engineer

XPO

- **Automated BigQuery DDL generation** via Python framework, facilitating 60+ object deployments through enhanced API metadata extraction and data type mapping.
- Developed IICS Taskflows for seamless integration of Salesforce and ServiceNow with BigQuery, optimizing data processing workflows.
- Implemented a BigQuery Command Center in Looker Studio to monitor slot utilization and utilized insights to reduce overcapacity and save **$120,000** annually through a **63%** reduction in slot renewal budget.
- Developed and managed **ETL/ELT data pipelines** between Oracle database and Google BigQuery

Google Cloud Platform Admin

XPO

- Oversaw comprehensive deployments leveraging **Cloud Composer (Airflow), Cloud Run, and Cloud Functions**.
- Employed **IAM roles** for GBQ Management and Terraform for service account creation and management.

EDUCATION

M.Tech

Indian Institute of Technology Bombay

07/2019 - 06/2021 India

GPA **9.58** / 10.0

- Developed a U-Net (ResNet backbone) based semantic segmentation model for land use & land cover classification on Copernicus/ESA satellite imagery
- Implemented YOLOv3 object detection pipeline to identify aircraft on airport runways during COVID mobility restrictions

KEY ACHIEVEMENTS

Cost Optimization

Achieved 63% BigQuery cost reduction, $120K+ annual savings through optimized data operations

Model Development

Improved ETA prediction accuracy by 15% through innovative ML modeling

Churn Revenue Capture

Successfully captured 50% of churn revenue using targeted machine learning strategies

INDEPENDENT PROJECTS

Email Automation:

- Built intelligent agentic email assistant capable of **auto-replying, drafting, and ignoring emails** using LLM-based decision logic
- Implemented a tool-based workflow with **SendGrid, SQLite, Pub/Sub, and OpenAI SDK**
- Integrated Pushover notifications and **ngrok-based webhook** handling
- Available at **mohit@aiguru360.in** for experience

Agentic Software Development

- Designed a **multi-agent orchestration** system (Manager + Backend + Frontend + QA agents) using **CREW framework**
- Incorporated **Docker-based execution validation** to verify generated code
- Demonstrated end-to-end autonomous software generation workflow

Resume Chatbot

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- Built a **Retrieval-Augmented Generation** (RAG) powered digital persona to answer career-related queries using **FAISS** and **MiniLM-L6-v2**
- Developed **conversational memory** to maintain multi-turn context and improve response coherence

SKILLS

GCP | SQL | SageMaker | AWS | LangGraph | Vertex AI | ETL | PySpark | Airflow | Docker | VectorDB | Deep Learning | Asyncio | CREW | Google ADK | PyTorch | TensorFlow | NLP | Agentic AI | LLM